RECOMMENDED TAG

Research

Verify It: Writing joins with mySQL

Why this type: This topic is a technique or procedure with a right and wrong way to do it, and students work from the dataset, and the planted error is caught by running it and seeing the result is wrong.

Research means participants verify AI's fluent, confident output against a real anchor and leave with a conclusion plus its evidence trail.

LENGTH	TEAMS	AI VISIBILITY
50 minutes · 5 phases	4 students, 4 roles	Embedded (Lane 2 — in class)

DOMINANT LITERACY	DOMINANT PILLAR
Critiquing AI and Recognizing AI	Pillar 3 — Critical Evaluation

READ THESE FIRST

PROTECT THE CLEANING STEP

Cleaning is where the judgment lives. Give it five explicit minutes and do not hand teams a pre-cleaned file — the session gets thinner the tidier the data is.

WORKFORCE-READINESS TEST

This is new territory. These students are still workforce-bound, so substitute close-to-home experience: ask what they have seen as a customer, a user, a student, or an employee rather than as a practitioner, and keep the transfer question pointed at the job they are heading toward.

How the four tags scored

Your content points most strongly at one tag. Subject never votes here; topic kind, material and the way the planted error gets caught do.

Research	10.0
Simulate	2.0
Create	1.5
Compete	-4.0

THE RUNNER-UP — SIMULATE

Tag it Simulate if you want students building something a colleague could walk through — depth of understanding becomes visible in the build.

TWO RULES THAT OVERRIDE ANY SCORE

No rival and no clock, no Compete. A countdown bolted onto a verification activity is decoration — if the audit survives without the timer, it is Research. **And if the artifact is for someone else to move through, it is Simulate, not Create**: a Create artifact is for the maker, a Simulate artifact is for a learner.

NOT THE TAG YOU WANT?

Change it

You know the class. Switching is allowed — the tool will tell you what the switch obligates you to add rather than quietly producing something thin.

Create

participants use AI to make something tangible

Research ✓ CURRENT

participants verify AI's fluent, confident output against a real anchor and leave with a conclusion plus its evidence trail

Compete

participants go head to head with a rival, or race a clock, with AI driving the pace and the score

Simulate

participants build an immersive experience that teaches someone else

THE ACTIVITY

What students actually do

AI produces something for Writing joins with mySQL that looks right. Teams run it against the dataset and find out that it is not.

How it runs

1. Teams start with the raw data on Writing joins with mySQL — working out what it actually measures and what has to be cleaned before it can be used at all.
2. AI returns a confident, clean-looking answer in seconds. It executes without error and is wrong in a specific way.
3. Running it is the verification. The gap between "it ran" and "it is right" is the whole lesson, and it only lands by doing it.

They leave with

A verified result on Writing joins with mySQL: what AI produced, what happened when it was run, and the specific failure that only execution exposed.

NOT THE ONE YOU PICTURED?

Three activities carry the Research tag, one per anchor. They teach the same skill, run on the same phases and timings, and hit the same wow and failure moments — only what students verify or make changes. The first ones are marked as a natural fit for the dataset; the other will work but need reframing.

Execute and Verify ✓ CHOSEN

AI produces something for Writing joins with mySQL that looks right. Teams run it against the dataset and find out that it is not.

Anchor: the executed result — what actually happens when you run it

The Data FITS THE DATASET

Teams clean and interrogate the dataset on Writing joins with mySQL to find out whether AI's finding actually holds. The cleaning is not preparation for the work; it is the work.

Anchor: the dataset itself — cleaning it is the verification

The Source FITS THE DATASET

Every claim AI makes about Writing joins with mySQL gets traced back through the dataset to a primary, verifiable origin — or marked as untraceable.

Anchor: the primary source each claim traces back to

AI as Investigator FITS THE DATASET

Teams turn AI loose on the dataset to hunt for a flaw planted in it — AI as the investigative instrument, not the thing being doubted.

Anchor: the target being investigated, with a flaw planted in it

Your Expertise

AI produces something fluent and subtly wrong about Writing joins with mySQL, and the only detection tool in the room is what students already know from the dataset.

Anchor: what the participant already knows about the field

CONCEPT CARD

The session, in one page

COURSE

CIS-255 · Analytics & Data

TOPIC

Writing joins with mySQL

WHAT HAPPENS NOW

pre-class lecture, practicing problem sets to introduce the topic

TOPIC KIND

A technique or procedure with a right and wrong way to do it

RAW MATERIAL

A dataset, results, or records. Phase 1 opens on it: Open the dataset. Before anything else, name what each field actually measures, what is missing or miscoded, and what you would have to clean before trusting one conclusion from it.

TRANSFERABLE SKILL

Verifying AI's confident output against something real — and knowing that fluency is not evidence.

THE ANCHOR

the executed result — what actually happens when you run it. Research runs on verification, not open-ended research — without a real anchor the session collapses into a summary.

THE PLANTED ERROR

They pick the wrong Join or join on the wrong field Seed this deliberately. “AI is wrong in exactly this way” builds; “AI might be wrong” does not.

ARTIFACT

A verified result on Writing joins with mySQL: what AI produced, what happened when it was run, and the specific failure that only execution exposed. A conclusion plus its trail. Never a summary.

WOW MOMENT

The AI answers in seconds, fluently and with total authority — an evening of work, compressed. That fluency is the wow, and it is the same fluency that turns out to be wrong. One lesson, delivered from both directions.

FAILURE BY DESIGN

The error is planted, not hoped for: AI is wrong in exactly one specified way, and the way is chosen in advance. Students find it by verifying against the anchor, never by being told. If nobody has it by the halfway mark, ask the room one question: which of these would you sign your name to?

PLANT THIS SPECIFIC ERROR

They pick the wrong Join or join on the wrong field

HUMAN CONTRIBUTION

The judgment call on evidence: deciding whether the anchor actually supports the claim attached to it. Only the student makes that call, and the artifact cannot be finished without it.

ENTRY POINT

Required: a laptop and the topic you already teach. Not required: any coding, any prior AI experience, any paid tool or account. Students use an institutionally approved AI tool of their choice.

SHARE FORMAT

State the conclusion and its evidence trail, then the one thing AI was confidently wrong about and how the anchor caught it. Works in a virtual classroom: screen-share plus one line in chat.

PILLAR 2 — SAFE USE

Approved tools only. No real patient, client, student or employee data goes into any AI tool. If a student wants to use a non-approved tool, they pair with someone who is on an approved one.

PILLAR 4 — ATTRIBUTION

The artifact carries one line: what AI produced, and what the team decided. Reflection question 3 asks for it out loud, in every version of this session.

Phases

#	PHASE	MIN	WHAT STUDENTS DO	UNLOCK WORD
1	**Set the anchor**	8	Open the dataset. Before anything else, name what each field actually measures, what is missing or miscoded, and what you would have to clean before trusting one conclusion from it.	—
2	**Let AI run**	10	Get AI's answer and write down what it claims before you run anything, then run it. Record what actually comes back, not what should have.	**sweep**
3	**Verify against the anchor**	14	Find where output and claim diverge. Prove the divergence, do not assert it.	**audit**
4	**Build the evidence trail**	10	Write the verified result: the claim, the run, the failure, the correction.	**trail**
5	**Present and reflect**	8	Show the moment it ran clean and was still wrong.	**present**

Roles — teams of 4

1. **Anchor keeper** — owns the written anchor and the claim under test, in one sentence each
2. **Prompter** — owns the prompt log: every prompt sent, in order
3. **Verifier** — owns the verdict on each claim and the evidence behind it
4. **Reporter** — owns the finished artifact: conclusion plus trail

AI skill moments — what to watch for

WHEN IT HAPPENS	THE SKILL	WHAT YOU DO
Phase 2, when a team accepts fluent output without an anchor behind it	**Prompting for evidence rather than assertion**	Ask them to re-prompt for what that specific sentence rests on. Let the room hear what comes back.
Phase 3, the first time the anchor contradicts the claim	**Verifying against something real, not against plausibility**	Put it on the projector. This is the highest-value sixty seconds in the session.
Phase 3, when a team treats a confident tone as a credential	**Separating fluency from accuracy**	Ask: how sure does it sound, and how sure should you be? Name the gap out loud.

Reflection

1. Where was the AI confident and wrong, and what made you check that one?
2. What is the difference between how fluent it sounded and how reliable it was?
3. Name what the AI produced and what you decided. Which of the two mattered more to how this turned out?
4. Where in your own work or study, in the next two weeks, could you use this exact move?

> *Closing stem, everyone completes it: "The most useful thing I learned about working with AI today was ___, and I could use this when ___."*

SAY THIS AT THE CLOSE

Knowing how to go about the process correctly

Pacing

If time runs short: halve the "Build the evidence trail" phase and cut sharing to three volunteers, everyone else submits one line. Never cut phase 2 ("Let AI run") or the reflection — the wow moment and the skill live there.

If energy is high: teams trade artifacts and try to break each other's conclusion against the same anchor — a five-minute adversarial pass.

HAND-OFF

Paste-ready build prompt

Copy this into your AI tool of choice to draft the full session, or send it to the AI Experience team as a finished brief. Every answer you gave is already in it.

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You are designing an SNHU AI Labs session (a "Lab"). Use the SNHU AI Experience framework skill:
work through the thirteen components, keep all participant-facing language tool-agnostic (never name
a specific AI tool), and deliver the finished session as one self-contained HTML file with a mode
chooser, projector stage and participant workbook, per the framework's design system.

This concept came out of a faculty intake. Treat the answers below as fixed inputs and do not re-ask
them.

== FACULTY INTAKE ==
Course: CIS-255
Topic / unit: Writing joins with mySQL
What happens in class now: pre-class lecture, practicing problem sets to introduce the topic
Subject: Analytics & Data — write the session in this field's language, using its own nouns (a
dataset, a model, a dashboard, a significance test, a data dictionary). Subject does not change the
tag.
Topic kind: A technique or procedure with a right and wrong way to do it — the activity language
must suit this kind of subject.
Raw material students work from: A dataset, results, or records (referred to throughout as "the
dataset")
Session length: 50 minutes | Class size: 12-25 | Teams of 4
Students bring lived/professional experience: no
Competing: No — the work rewards care over speed and there is no rival
How a student catches the planted error: By running it and seeing the result is wrong
Faculty's worry about AI here: thinking
What goes wrong most in student work now: it is wrong and unchecked — the failure by design should
bite exactly here.
Expert-only mistake to plant as the designed failure: They pick the wrong Join or join on the wrong
field
What students should remember a year later: Knowing how to go about the process correctly

== CONFIRMED DESIGN DECISIONS ==
Experience type: Research (participants verify AI's fluent, confident output against a real anchor
and leave with a conclusion plus its evidence trail)
ACTIVITY (chosen by the faculty member from three options for this type) — "Execute and Verify": AI
produces something for Writing joins with mySQL that looks right. Teams run it against the dataset
and find out that it is not.
Activity fits the raw material naturally: yes
RESEARCH ENGINE — the three slots, all non-negotiable:
  1. Anchor: the executed result — what actually happens when you run it
  2. Planted error: They pick the wrong Join or join on the wrong field
  3. Evidence artifact: A verified result on Writing joins with mySQL: what AI produced, what
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happened when it was run, and the specific failure that only execution exposed. (a conclusion plus its trail — never a summary)
Research pattern (by anchor): Execute and Verify. The center of this type is verification, not open-ended research.
Do not attach a clock as a mechanic: if the audit survives without the timer, it is Research, not Compete.
How the activity runs:
(1) Teams start with the raw data on Writing joins with mySQL — working out what it actually measures and what has to be cleaned before it can be used at all.
(2) AI returns a confident, clean-looking answer in seconds. It executes without error and is wrong in a specific way.
(3) Running it is the verification. The gap between “it ran” and “it is right” is the whole lesson, and it only lands by doing it.
Runner-up considered: Simulate
AI visibility: embedded (this drops into an existing class, so the AI skill surfaces mainly in reflection) — Lane 2.
Dominant literacy category: Critiquing AI and Recognizing AI | Secondary: Using AI
Dominant pillar: Pillar 3 — Critical Evaluation | Secondary: Pillar 4 — Transparency and Attribution
Baseline pillars, required: Pillar 2 Safe and Compliant Use — institutionally approved tools only, no real patient/client/student data entered; Pillar 4 Transparency and Attribution — the artifact carries a line stating what AI produced and what the team decided.
Transferable AI skill: Verifying AI's confident output against something real — and knowing that fluency is not evidence.
Artifact: A verified result on Writing joins with mySQL: what AI produced, what happened when it was run, and the specific failure that only execution exposed.
Wow moment: The AI answers in seconds, fluently and with total authority — an evening of work, compressed. That fluency is the wow, and it is the same fluency that turns out to be wrong. One lesson, delivered from both directions.
Failure by design: The error is planted, not hoped for: AI is wrong in exactly one specified way, and the way is chosen in advance. Students find it by verifying against the anchor, never by being told. If nobody has it by the halfway mark, ask the room one question: which of these would you sign your name to?
Tie-in: Faculty's stated worry is students outsourcing the thinking. The role split is the countermeasure: the AI-facing role never makes the final call — a person does, on the record.
Human contribution moment: The judgment call on evidence: deciding whether the anchor actually supports the claim attached to it. Only the student makes that call, and the artifact cannot be finished without it.
Share format: State the conclusion and its evidence trail, then the one thing AI was confidently wrong about and how the anchor caught it. Must work in a virtual classroom: screen-share plus one line in chat.
Entry point — required: a laptop and the topic already being taught. Not required: any coding, any prior AI experience, any paid tool or account. Participants use an institutionally approved AI tool of their choice.
Pacing flexibility — if short: halve the "Build the evidence trail" phase and cut sharing to three volunteers, everyone else submits one line. Never cut phase 2 ("Let AI run") or the reflection. If energy is high: teams trade artifacts and try to break each other's conclusion against the same anchor — a five-minute adversarial pass.

== PHASES (this brief is written for 50 min; 50 / 75 / 90 columns given so the design system's 50-75 toggle can be built directly) ==

1. Set the anchor — 50min: 8 | 75min: 12 | 90min: 14 — facilitator function: Name the real thing this session verifies against.
    participant task: Open the dataset. Before anything else, name what each field actually measures, what is missing or miscoded, and what you would have to clean before trusting one conclusion from it.   [no unlock word; opening phase]
2. Let AI run — 50min: 10 | 75min: 15 | 90min: 18 — facilitator function: Get AI's fluent, confident output — before checking anything.
    participant task: Get AI's answer and write down what it claims before you run anything, then run it. Record what actually comes back, not what should have.   [unlock word: sweep]
3. Verify against the anchor — 50min: 14 | 75min: 20 | 90min: 24 — facilitator function: Check every claim against the anchor. Mark what fails.
    participant task: Find where output and claim diverge. Prove the divergence, do not assert it. [unlock word: audit]
4. Build the evidence trail — 50min: 10 | 75min: 16 | 90min: 20 — facilitator function: Conclusion plus its trail — never a summary.
    participant task: Write the verified result: the claim, the run, the failure, the correction. [unlock word: trail]
5. Present and reflect — 50min: 8 | 75min: 12 | 90min: 14 — facilitator function: Conclusion, evidence, and the moment AI was confidently wrong.
    participant task: Show the moment it ran clean and was still wrong.   [unlock word: present]

== AI SKILL MOMENTS (what the facilitator watches for) ==
1. Phase 2, when a team accepts fluent output without an anchor behind it -> skill: Prompting for evidence rather than assertion -> facilitator move: Ask them to re-prompt for what that specific sentence rests on. Let the room hear what comes back.
2. Phase 3, the first time the anchor contradicts the claim -> skill: Verifying against something real, not against plausibility -> facilitator move: Put it on the projector. This is the highest-value sixty seconds in the session.
3. Phase 3, when a team treats a confident tone as a credential -> skill: Separating fluency from accuracy -> facilitator move: Ask: how sure does it sound, and how sure should you be? Name the gap out loud.

== ROLES (teams of 4, each with a named deliverable) ==
1. Anchor keeper — owns the written anchor and the claim under test, in one sentence each
2. Prompter — owns the prompt log: every prompt sent, in order
3. Verifier — owns the verdict on each claim and the evidence behind it
4. Reporter — owns the finished artifact: conclusion plus trail

== REFLECTION ==
1. Where was the AI confident and wrong, and what made you check that one?
2. What is the difference between how fluent it sounded and how reliable it was?
3. Name what the AI produced and what you decided. Which of the two mattered more to how this turned out?
4. Where in your own work or study, in the next two weeks, could you use this exact move?
Closing stem, every participant completes aloud or in writing: "The most useful thing I learned about working with AI today was ___, and I could use this when ___."

== BUILD NOTES ==
Session slug: writing-joins-with (broadcast channel snhu-writing-joins-with-room, localStorage key snhu-writing-joins-with-proto)

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Timing map: use the 50 and 75 columns above as the t50/t75 proportional map required by the design
system. The 90-minute column is faculty-facing planning only — do not add a third toggle.
Run the framework's quality checklist before you finish, including the Research-specific checks and
both baseline pillar checks.
Then produce: (A) the tool-agnostic session HTML, (B) a tool-specific version only if a named tool's
capability would meaningfully change the activity, and (C) the build instructions that reproduce it.
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